Camera Calibration (USB)

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1. Install Plugins

Install Image Calibration Tool

```
sudo apt install ros-${ROS_DISTRO}-image-pipeline -y
```

```
jetson@ubuntu:~$ sudo apt install ros-${ROS_DISTRO}-image-pipeline -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ros-jazzy-image-pipeline is already the newest version (5.0.11-1noble.20260124.2
02428).
The following packages were automatically installed and are no longer required:
  libdjvulibre-dev liblqr-1-0-dev libmagick++-6-headers libmagick++-6.q16-9t64
  libmagickcore-6-arch-config libmagickcore-6-headers libmagickwand-6-headers
  librsvg2-dev
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 26 not upgraded.
jetson@ubuntu:~$
```

Install Official USB Camera Tool

```
sudo apt install ros-${ROS_DISTRO}-usb-cam -y
```

```
jetson@ubuntu:~$ sudo apt install ros-${ROS_DISTRO}-usb-cam -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ros-jazzy-usb-cam is already the newest version (0.8.1-1noble.20260124.191846).
The following packages were automatically installed and are no longer required:
  libdjvulibre-dev liblqr-1-0-dev libmagick++-6-headers libmagick++-6.q16-9t64
  libmagickcore-6-arch-config libmagickcore-6-headers libmagickwand-6-headers
  librsvg2-dev
```

2. Start the Camera

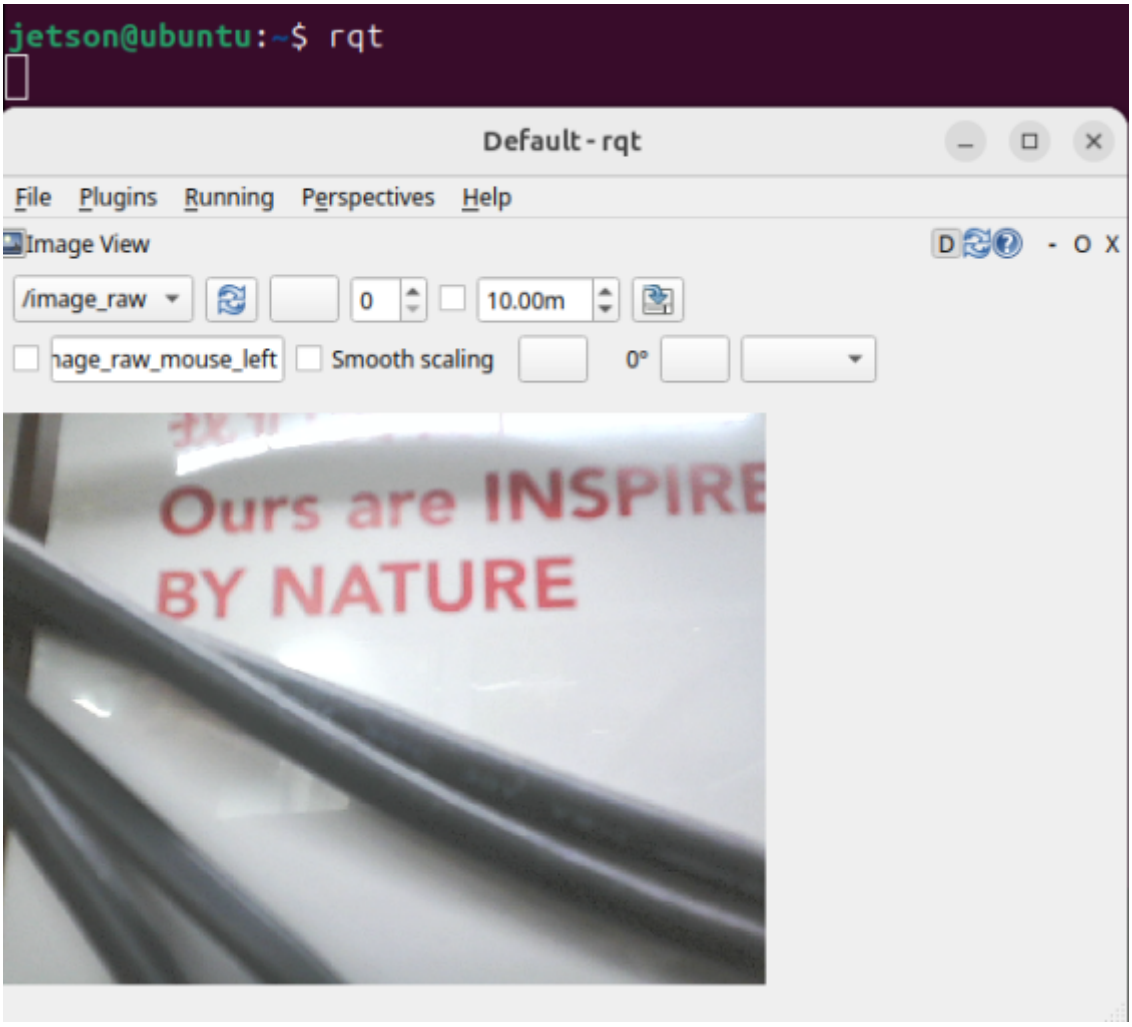
Restart the terminal, re-enable the ROS environment, and then run the following command to start the USB camera node.

```
ros2 run usb_cam usb_cam_node_exe
```

```
jetson@ubuntu:~$ ros2 run usb_cam usb_cam_node_exe
[INFO] [1773373724.322634368] [usb_cam]: camera_name value: default_cam
[WARN] [1773373724.322719979] [usb_cam]: framerate: 30.000000
[INFO] [1773373724.324599907] [usb_cam]: using default calibration URL
[INFO] [1773373724.324641527] [usb_cam]: camera calibration URL: file:///home/je
tson/.ros/camera_info/default_cam.yaml
[WARN] [1773373724.325029907] [usb_cam]: [default_cam] does not match test_camer
a in file /home/jetson/.ros/camera_info/default_cam.yaml
[INFO] [1773373724.393743162] [usb_cam]: Starting 'default_cam' (/dev/video0) at
 640x480 via mmap (yuyv) at 30 FPS
[swscaler @ 0xaaaae8283780] No accelerated colorspace conversion found from yuv4
22p to rgb24.
```

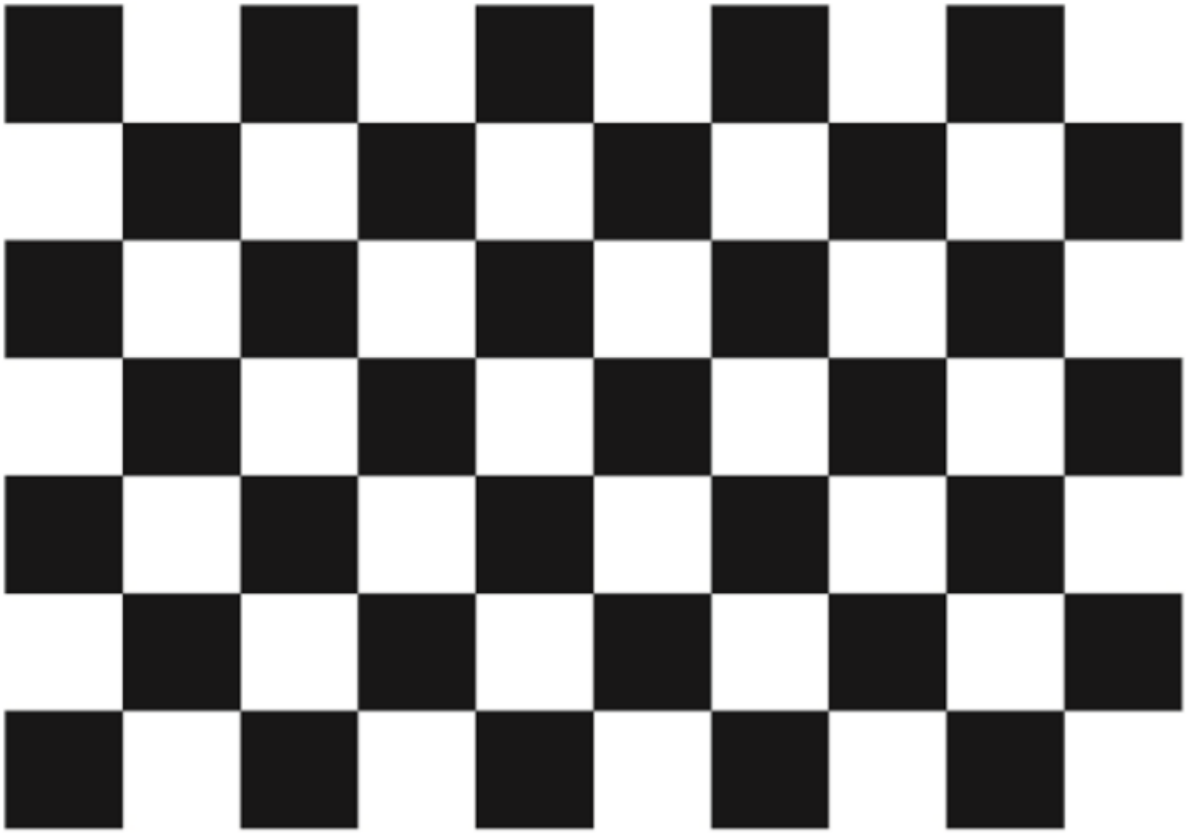
Open another terminal and launch rqt:

```
rqt
```



3. Image Calibration

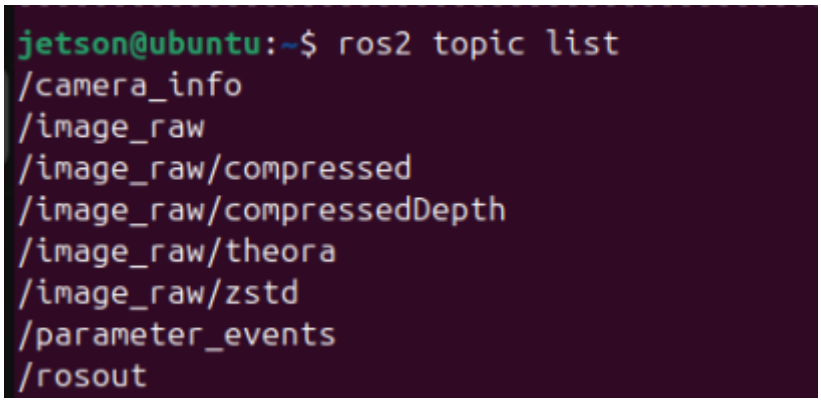
3.1. Calibration Board



7×10 | Size: 20mm

3.2. Viewing Topics and Services

```
ros2 topic list
ros2 service list
```



```
jetson@ubuntu:~$ ros2 service list
/rqt_gui_cpp_node_16653/describe_parameters
/rqt_gui_cpp_node_16653/get_parameter_types
/rqt_gui_cpp_node_16653/get_parameters
/rqt_gui_cpp_node_16653/get_type_description
/rqt_gui_cpp_node_16653/list_parameters
/rqt_gui_cpp_node_16653/set_parameters
/rqt_gui_cpp_node_16653/set_parameters_atomically
/set_capture
/usb_cam/describe_parameters
/usb_cam/get_parameter_types
/usb_cam/get_parameters
/usb_cam/get_type_description
/usb_cam/list_parameters
/usb_cam/set_camera_info
/usb_cam/set_parameters
/usb_cam/set_parameters_atomically
```

3.3. Image Calibration

Launch the Image Calibration Tool

Launch the ROS 2 camera calibration tool, `camera_calibration`, to perform the camera calibration. Note: The camera must be started *before* running the calibration tool.

```
ros2 run camera_calibration cameracalibrator --size 9x6 --square 0.02 --ros-args
--remap /image:=/image_raw --remap
camera/set_camera_info:=usb_cam/set_camera_info
```

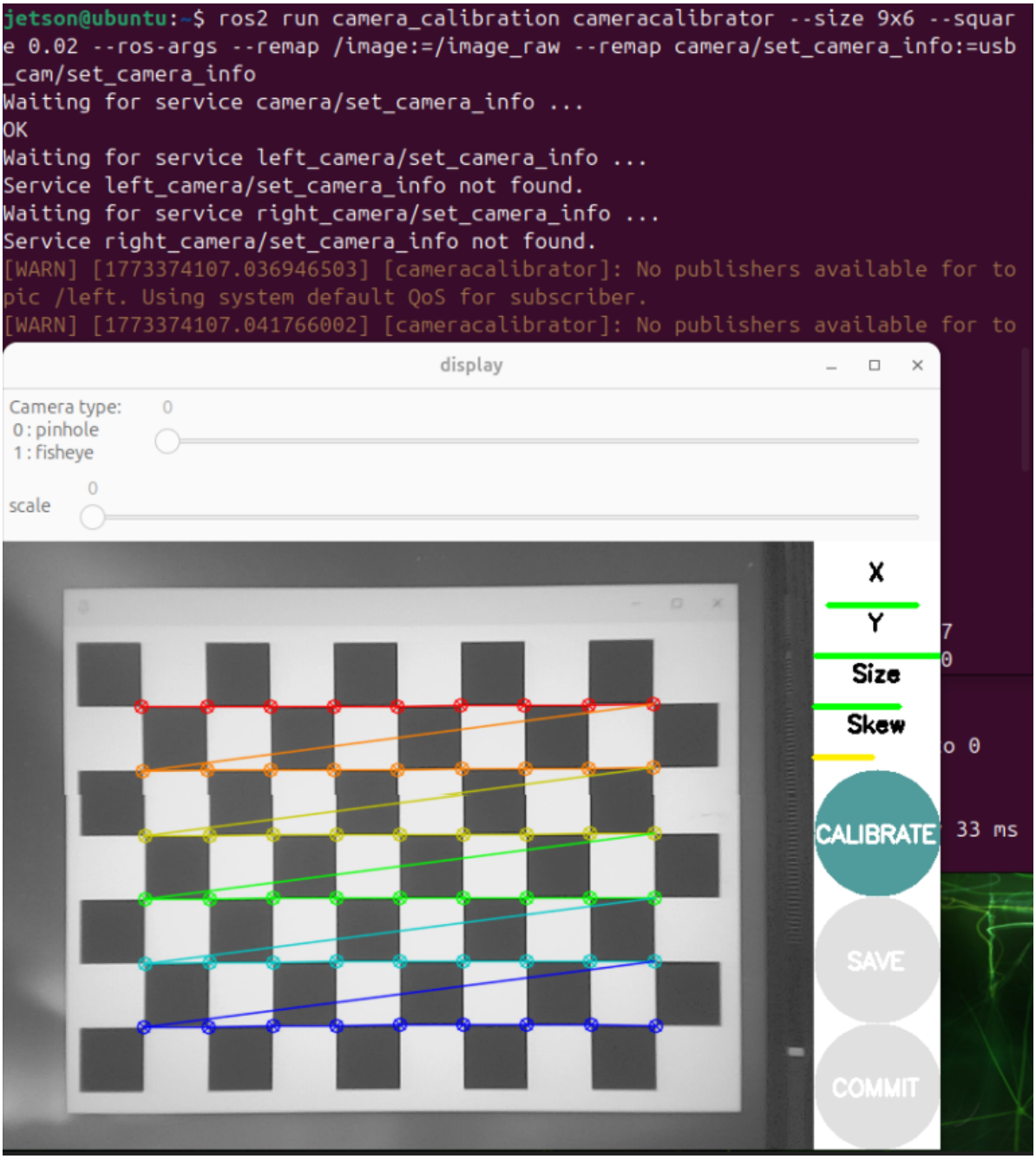
> **Parameter Explanation**

`--size`: The number of *internal* corners on the chessboard pattern → 9 rows by 6 columns of internal corners.

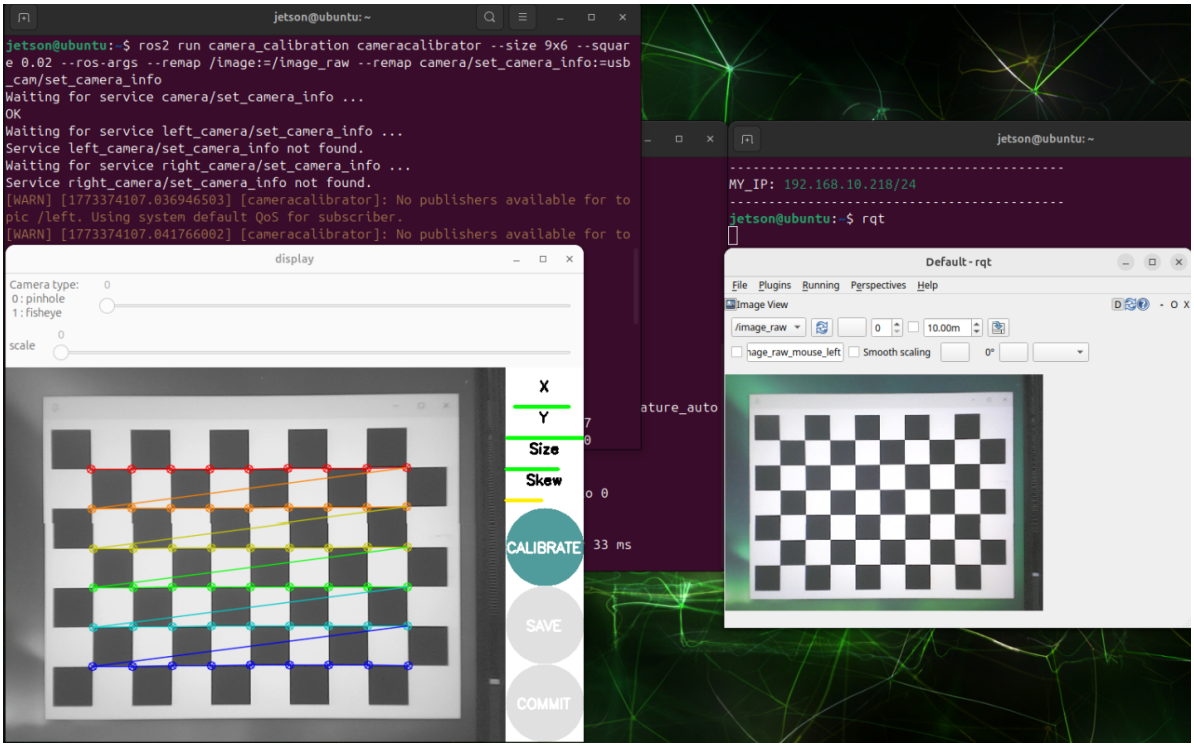
`--square`: The side length of each square on the chessboard pattern → 0.02 meters

`--ros-args`: Pass ROS arguments

`--remap`: Topic remapping → `/image_raw` serves as the image input stream



Calibration Process



> **Parameter Descriptions**

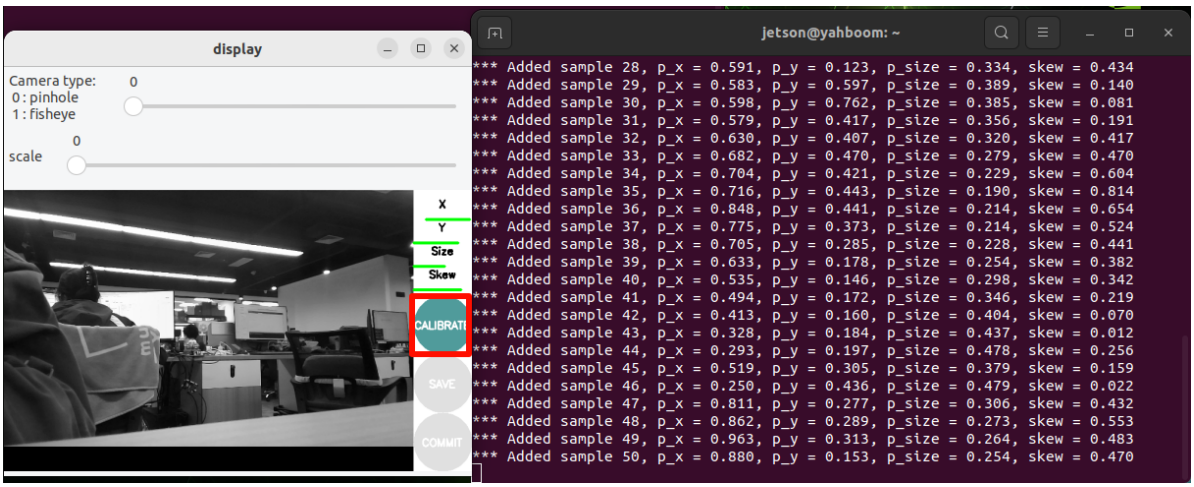
`X`: Move the chessboard left and right within the camera's field of view

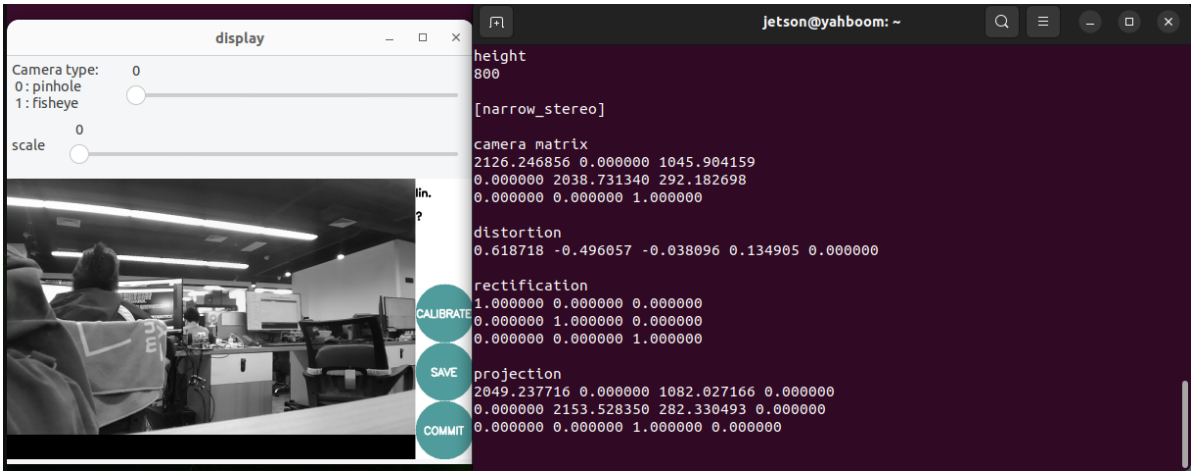
`Y`: Move the chessboard up and down within the camera's field of view

`Size`: Move the chessboard forward and backward within the camera's field of view

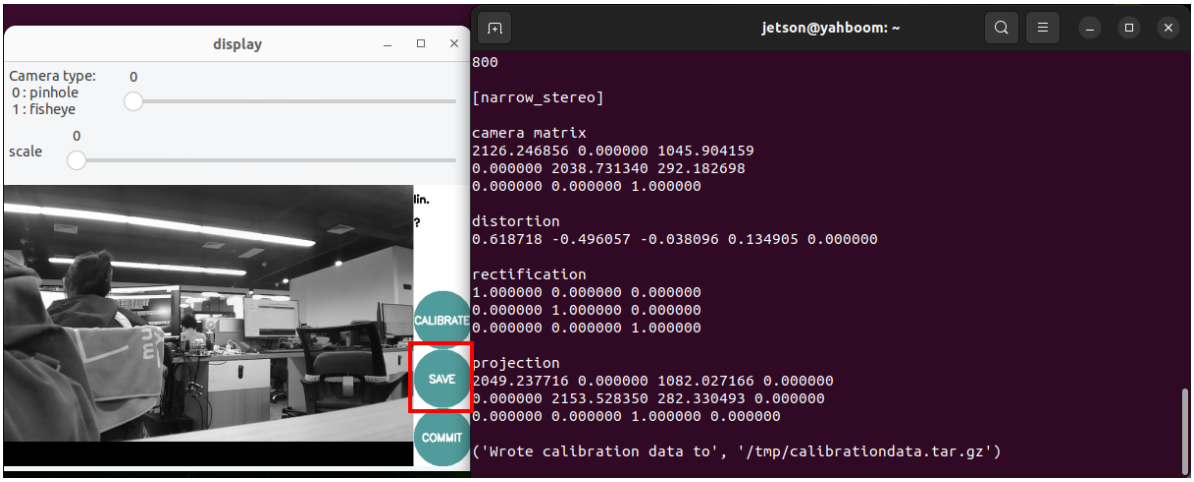
`Skew`: Tilt and rotate the chessboard within the camera's field of view

When `X`, `Y`, `Size`, and `Skew` turn green, you can click `CALIBRATE` to compute the camera's intrinsic parameters. Note: The more images used for calibration, the longer the process will take, and there is a possibility that the program may freeze.





Once calibration is complete, click `SAVE` to save the results!



Close the program after the results have been successfully saved! **Calibration Results**

Default save location and filename for the calibration results:

```
/tmp/calibrationdata.tar.gz
```

Copy and extract the calibration results:

```
sudo mv /tmp/calibrationdata.tar.gz ~
mkdir calibrationdata
tar -xzvf calibrationdata.tar.gz -C ~/calibrationdata
```

The extracted files include the calibration `*.png`, `ost.yaml`, and `ost.txt` files. We will primarily copy the data contained within the `ost.yaml` file into the `camera_usb.yaml` file.

> **ost.yaml**

Replace the corresponding data in the `camera_usb.yaml` file with the values for `image_width`, `image_height`, the `data` field within `camera_matrix`, the `data` field within `projection_matrix`, and the `data` field within `distortion_coefficients`.

```
image_width: 1280
image_height: 800
camera_name: narrow_stereo
camera_matrix:
  rows: 3
  cols: 3
```

```
  data: [2126.24686,    0.     , 1045.90416,
            0.     , 2038.73134,  292.1827 ,
            0.     ,    0.     ,    1.     ]
distortion_model: plumb_bob
distortion_coefficients:
  rows: 1
  cols: 5
  data: [0.618718, -0.496057, -0.038096, 0.134905, 0.000000]
rectification_matrix:
  rows: 3
  cols: 3
  data: [1., 0., 0.,
         0., 1., 0.,
         0., 0., 1.]
projection_matrix:
  rows: 3
  cols: 4
  data: [2049.23772,    0.     , 1082.02717,    0.     ,
            0.     , 2153.52835,  282.33049,    0.     ,
            0.     ,    0.     ,    1.     ,    0.     ]
```

> **camera_usb.yaml**

Modified File:

```
%YAML:1.0
---
image_width: 1920
image_height: 1080
camera_name: camera
camera_matrix: !!opencv-matrix
  rows: 3
  cols: 3
  dt: d
  data: [2126.24686,    0.     , 1045.90416,
            0.     , 2038.73134,  292.1827 ,
            0.     ,    0.     ,    1.     ]
distortion_model: plumb_bob
distortion_coefficients: !!opencv-matrix
  rows: 1
  cols: 5
  dt: d
  data: [0.618718, -0.496057, -0.038096, 0.134905, 0.000000]
rectification_matrix:
  rows: 3
  cols: 3
  data: [1., 0., 0.,
         0., 1., 0.,
         0., 0., 1.]
projection_matrix:
  rows: 3
  cols: 4
  data: [2049.23772,    0.     , 1082.02717,    0.     ,
            0.     , 2153.52835,  282.33049,    0.     ,
            0.     ,    0.     ,    1.     ,    0.     ]
```